# Module 3: Mass storage Interfaces

Mass storage Interfaces: The IDE interface, The SCSI interface, Comparisons. Magnetic Storage devices: Magnetic Storage, Hard disk drive. Optical storage Devices: Optical storage Media, DVD-ROM drives, Recordable drives. SSD.

## Mass storage Interfaces

A mass storage device (MSD) is any storage device that makes it possible to store and port large amounts of data across computers, servers and within an IT environment. MSDs are portable storage media that provide a storage interface that can be both internal and external to the computer. Eg: HDD,CD-ROM,DVD,USB

In order for these devices to communicate with the system, they must be connected to an interface. The two primary interface types used in PCs today are the Integrated Drive Electronics (IDE) interface and the Small Computer System Interface (SCSI).

## The IDE interface

**IDE Interface**. The primary **interface** used to connect a hard disk drive to a PC is typically called **IDE** ( Integrated Drive Electronics). IDE (Integrated Drive Electronics) is a standard electronic interface used between a computer motherboard's data paths or bus and the computer's disk storage devices. The IDE interface is based on the IBM PC Industry Standard Architecture (ISA) 16-bit bus standard, but it is also used in computers that use other bus standards. The ANSI name for IDE is Advanced Technology Attachment (ATA).

There are two ATA interface types: Parallel ATA (PATA) and Serial ATA (SATA).

## ATA standards

**ATA-1**

**ATA-2**

**ATA-3**

**ATA-4**

**ATA-5**

**ATA-6**

**ATA-7**

## The ATA I/O Cable

Desktop systems use a ribbon cable with 40 wires and three identical female pin connectors, technically called Insulation-Displacement Connectors (IDCs), that attach to two storage devices and the bus connector on the motherboard. In some cases, laptop and other portable systems use a 44-pin cable, the extra four pins of which are used to carry power to the drive.

An IDE cable has a red stripe along one edge, like you see below. It's that side of the cable that usually refers to the first pin. The connectors on the ATA cable are usually keyed to prevent you from inserting them upside down, using two methods. Pin 2o in all three connectors on the cable is plugged, which lines up with a missing pin in the sockets, and there is also a detent on each connector that matches a cutout on the socket , ensuring proper alignment of the pins.

The blue connector is used to connect the cable to the mainboard of the computer/device, while the black one is used to connect the drive. IDE cables usually have two drive connectors: a "master" connector (at the end of the cable) and a "slave" connector (in the middle of the cable, closer to the "master" connector).

### Master slave configuration

To allow for two drives on the same cable, IDE uses a special configuration called master and slave.

This configuration allows one drive's controller to tell the other drive when it can transfer data to or from the computer.

### Setting Master / slave Jumpers

A jumper on the back of the drive next to the IDE connector must be set in the correct position to identify the drive as the master drive. The slave drive must have either the master jumper removed or a special slave jumper set, depending on the drive. Also, the slave drive is attached to the connector near the middle of the IDE ribbon cable. Each drive's controller board looks at the jumper setting to determine whether it is a slave or a master. This tells them how to perform. Every drive is capable of being either slave or master when you receive it from the manufacturer. If only one drive is installed, it should always be the master drive.

Many drives feature an option called Cable Select (CS). With the correct type of IDE ribbon cable, these drives can be auto configured as master or slave. CS works like this: A jumper on each drive is set to the CS option. The cable itself is just like a normal IDE cable except for one difference -- Pin 28 only connects to the master drive connector. When your computer is powered up, the IDE interface sends a signal along the wire for Pin 28. Only the drive attached to the master connector receives the signal. That drive then configures itself as the master drive. Since the other drive received no signal, it defaults to slave mode.

### Data Transfer Modes

The performance of a particular HDD on an IDE interface is largely determined by the data transfer mode used to transmit the data between the computer and the drive.

- Programmed Input/Output (PIO)
- Direct Memory Access
- Ultra DMA

# The SCSI Interface

The Small Computer System Interface is the second most popular PC drive interface in use today. Generally considered to be more of a high-end solution than IDE.SCSI is more expensive, but it supports more devices and more types of devices. SCSI is also better suited to network environments in which many users are accessing shared drives simultaneously.

**Comparisons**

- Unlike IDE, the SCSI interface generally is not built into PCs, although some motherboards with integrated SCSI adapters do exist. In most cases users have to purchase a SCSI host adapter , which forms an independent bus within the computer.
- IDE was originally designed to be a hard drive interface, and support for other types of devices was patched on later. SCSI, on the other hand, was designed from the start to support a wide variety of devices, including hard disk drives, CD-ROM drives, tape drives, cartridge drives, media changers and graphics scanners. SCSI host adapters have both internal and external connectors, so it can use the same adapter to connect both your hard drives and CD-ROM drives inside the computer and external devices to the bus.
- SCSI devices are somewhat similar to their IDE counterparts insofar as they have their controllers integrated into the unit.

# SCSI bus

The SCSI bus is a parallel bus, which comes in a number of variants. The oldest and most used is an 8 bit wide bus, with single-ended signals, carried on 50 wires. Modern designs also use 16 bit wide buses, with differential signals.SCSI bus not only has data lines, but also a number of control signals. A very elaborate protocol is part of the standard to allow multiple devices to share the bus in an efficient manner. SCSI is a bus that supports as many as 7 or 15 total devices. Multichannel adapters exist that can support up to 7 or 15 devices per channel. The SCSI host adapter might be installed as a PCI or as a PCI-Express card, or it might be built in to the motherboard.The host adapter functions as the gateway between the SCSI bus and the PC system bus. Each device on the bus has a SCSI controller chip built in. The SCSI host adapter does not talk directly with actual devices such as hard disks; instead, it talks to the SCSI interface controller that is built in to the drive.

## SCSI Standards

The standard that defines the SCSI interface seems to have undergone a rather simple development process.

- SCSI 1
- SCSI 2
- SCSI 3
- Serial SCSI

| Topic | ATA/IDE | SCSI |
| --- | --- | --- |
| Cost | Overall, IDE is a much cheaper solution. | Compared with IDE, SCSI is often more expensive to implement and support. |
| Expansion | IDE/EIDE allows 2 two devices per channel. Most computers have 2 channels. | SCSI is capable of supporting up to 7 or 15 devices. |
| Ease | IDE is commonly an easier product to set up than SCSI. | Configuring SCSI can be more difficult for most users compared to IDE. |
| Faster | Today, the latest IDE and SCSI drives running at the same RPM are very close. However, 10,000+ RPM drives are often only available for SCSI. | All the fastest drives are often available for SCSI first and in many cases 10,000+ RPM hard drives are only available as SCSI drives. |
| Resources | All motherboards today have an ATA/IDE interface and unless additional drives are needed no additional resources need to be taken. | Unlike IDE, SCSI requires an interface expansion card in most cases (unless the motherboard already has it). Adding any new hardware means more system resources are going to be required. |

# Magnetic Storage devices

In the **Magnetic storage devices**, all data are stored with using magnetized medium, and those types of data saved in that medium in the binary form like as 0 and 1. This magnetic storage has also non-volatile storage nature. Today's, mostly people are preferred to magnetic medium because on the **magnetic storage devices** can be performed read/write activities very easily. Magnetic storage devices have huge capacities for storing data that it's more attractive point. These storage devices are not more costly but their data accessing power is slow.

- Eg: Hard Disk, Floppy Disk, Magnetic Tape,

Magnetic Storage

Hard disks, floppy disks and tape drives as well as some cartridge drives, all store data using the principles of electromagnetism. In each case, the storage medium consists of a moving surface coated with particles that are capable of holding a magnetic charge. The particles are based on an iron oxide material. The surface can be rigid such as the aluminum or glass platters in a hard disk drive or flexible like the Mylar or other polymers used in floppy disk and tapes.

## Writing Data

In their original state, each of the particles on the storage medium has a magnetic charge, but the charges are disorganized. The magnetic field point in random directions and cancel each other out. When there is no cumulative magnetic charge on the surface of the medium, it contains no data. To record data onto the medium, a recording head passes close to the surface. The head is a U shaped conductor that carries an electric charge, which turns it into an electromagnet. When the points of the U are passed over the surface of the storage medium, the electromagnetic charge passes through the conductive surface of the medium more easily.

When the electric charge from the head encounters the magnetic particles on the recording medium, it aligns them all in the same direction, depending on the polarity of the charge running through the head. When the particles on the medium are aligned, the individual magnetic fields no longer cancel each other out and instead produce a cumulative charge called a flux. The drive is capable of changing the polarity of the charge running through the head very quickly. When the polarity of the charge in the head reverses, so does the direction of the particle alignment, producing a flux transition or flux reversal. As the medium moves in relation to the head, the head creates a pattern of flux transition using a code to represent binary bit values.

The medium is divided into units called bit cells or transition cells, each of which contains the flux transitions that represent one bit of data. Once the particles on the medium are aligned in a particular pattern, they remain that way until the head applies another charge, which changes the pattern. This is what makes a magnetic medium a permanent storage solution.

## Reading Data

When reading data from magnetic media, the head passes over the surface of the medium. The head generates a voltage pulse only when it passes over a flux transition.A transition from a positive to a negative charge registers as a negative voltage pulse, and a negative to positive transition registers as a positive voltage pulse. The drive’s controller amplifies the pulses and decodes them to reproduce the original data. IBM developed a new type of read technology called a magneto-resistive (MR) head. The MR head is separate from the write head, and functions by measuring the nonlinear resistance charge as a bias current passes through the head that is traveling over the surface of the disk. The drive circuitry monitors the current passing through the head; when the flux transitions cause increased resistance, the voltage changes in the same pattern as when the data was written to the medium. The difference here is that the signal is much more powerful and much easier to discern from the background noise.

## Magnetic Encoding Schemes

Digital information is a stream of ones and zeros. Hard disks store information in the form of magnetic pulses. In order for the PC's data to be stored on the hard disk, therefore, it must be converted to magnetic information. When it is read from the disk, it must be converted back to digital information. This work is done by the integrated controller built into the hard drive, in combination with sense and amplification circuits that are used to interpret the weak signals read from the platters themselves.

Magnetic information on the disk consists of a stream of magnetic fields. a magnet has two poles, north and south, and magnetic energy (called *flux*) flows from the north pole to the south pole. Information is stored on the hard disk by encoding information into a series of *magnetic fields*. This is done by placing the magnetic fields in one of two polarities: either so the north pole arrives before the south pole as the disk spins (N-S), or so the south pole arrives before the north (S-N).

**FM Encoding Scheme**:

- FM or Frequency Modulation was the original data-encoding scheme used for storing the data on the magnetic recording surface.
- This method of data encoding is also known as the "Single density recording"
- In this method, a clock signal is put with every data signal on the recording surface. This clock signal is used for synchronizing the read operation, as there will always be a clock signal, whether the data signal is there or not.
- In this FM method of data recording a 1 bit is stored as two pulses(one clock pulse and one data pulse), and a o bit is stored as a one pulse and one gap or no pulse.
- For example, a binary number 1011 will be stored as PP PN PP PP

**MFM Encoding Scheme**

- More data can be stored on the same surface or the data storage density can be increased, if the number of pulses required to store the data can be minimized.
- The MFM (modified frequency modulation) method of data storage, by reducing the number of pulses, is able to store more data without any data and synchronization number of pulses, is able to store more data without any data and synchronization loss.
- In MFM recording the 0s and 1s are encoded as given below
- 1 is always stored as no pulse, and a pulse(NP)
- 0, when preceded by another 0, is stored as a pulse, and no pulse(PN)
- 0, when preceded by a 1, is stored as two no pulses(NN)
- If you store 1001 on the disk surface using the MFM storage method, it would be stored as NP NN PN NP.

**RLL Encoding Scheme**

- The RLL is encoding or the run length limited encoding is the most common encoding scheme used in the hard disk storage.
- This encoding scheme can be more accurately called as 2,7 RLL encoding because in this scheme in a series or in a running length the minimum number of 0s next to each other is two, and the maximum number of 0s together can not be more than seven.

- The RLL encoding scheme can store 50 percent more information than MFM encoding scheme on a given surface and it can store three times as much information as the FM encoding scheme. The Run length Limited name comes from the minimum number (run Length) and maximum number (run Limit) of “no pulse” values allowed between two pulses.
- For the RLL encoding, an encoder/decoder (Endec) table is used to find the pulse signal to be used for different data bit groups. Endec table used by the IBM to convert bit information to the pulse signal is shown below.

| Data Bit | Pulse |
|---|---|
| 10 | NPNN |
| 11 | PNNN |
| 000 | NNNP |
| 010 | PNNP |
| 011 | NNPN |

For example, if you want to encode a byte 100011 to proper RLL pulse signal then the Bit 10 can be encoded as NPNN Bit 0011can be encoded as NNNNPNNN .

# Hard disk drive

A hard disk drive (sometimes abbreviated as a hard drive, HD, or HDD) is a non-volatile data storage device. It is usually installed internally in a computer, attached directly to the disk controller of the computer's motherboard. It contains one or more platters, housed inside of an air-sealed casing. Data is written to the platters using a magnetic head, which moves rapidly over them as they spin. Internal hard disks reside in a drive bay, connected to the motherboard using an ATA, SCSI, or SATA cable. They are powered by a connection to the computer's PSU (power supply unit).

### Track

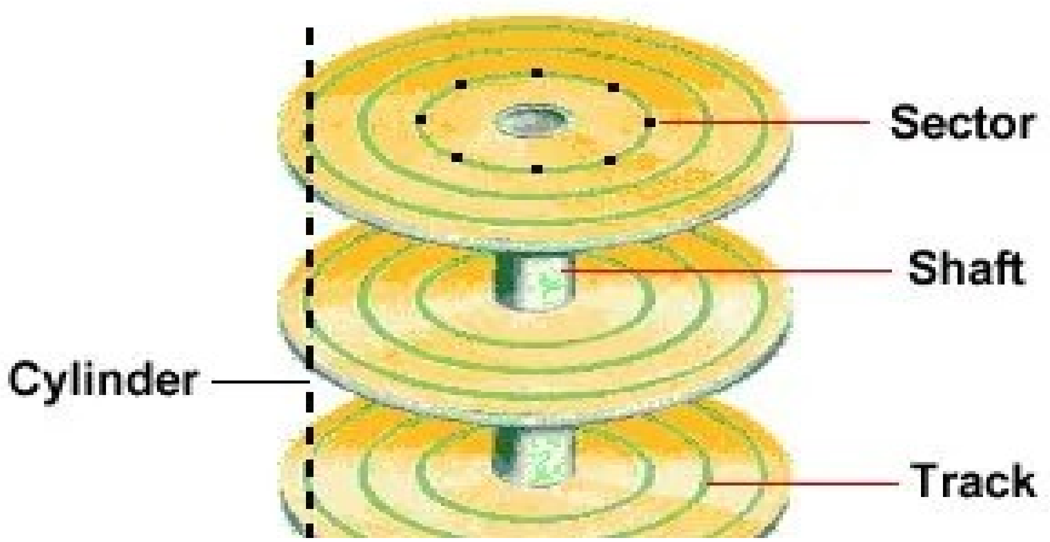
A **disk drive track** is a circular path on the surface of a **disk** or diskette on which information is magnetically recorded and from which recorded information is read.

### Sector

Each track is further broken down into smaller units called sectors. As sector is the basic unit of data storage on a hard disk. A single track typically can have thousands of sectors and each sector can hold more than 512 bytes of data. A few additional bytes are required for control structures and error detection and correction.

### Cylinder

The concept is concentric, hollow, cylindrical slices through the physical disks (platters), collecting the respective circular tracks aligned through the stack of platters. The number of cylinders of a disk drive exactly equals the number of tracks on a single surface in the drive. It comprises the same track number on each platter, spanning all such tracks across each platter surface that is able to store data.




# Hard Drive Components

- Platters
- Heads
- Head Actuator
- Spindle Motor
- Logic Board

### Platters

A hard disk driver consists of on one or more flat circular discs called platters on witch data is stored. Platters are made from a rigid material like aluminum or glass. The drive consists of a number of platters stacked on top of each other and threaded through a central spindle that is connected to the spindle motor, which spins the entire platter assembly. HDDs can have any number of platters.

**Platter Construction**

Hard disks are rigid platters, composed of a substrate and a magnetic medium. The substrate – the platter's base material – must be non-magnetic and capable of being machined to a smooth finish. It is made either of aluminum alloy or a mixture of glass and ceramic. To allow data storage, both sides of each platter are coated with a magnetic medium – formerly magnetic oxide, but now, almost exclusively, a layer of metal called a thin-film medium. This stores data in magnetic patterns, with each platter capable of storing a billion or so bits per square inch (bpsi) of platter surface.

## Heads

The heads in a HDD are responsible for writing data to and reading it from the magnetic media applied to the platter surfaces.Disk read/write heads are the small parts of a disk drive which move above the disk platter and transform the platter's magnetic field into electrical current (read the disk) or, vice versa, transform electrical current into magnetic field. The heads are mounted on an **actuator arm** in such a way as to place each head in contact with the platter surface it is assigned to read and write. When the drive is not running, the heads are in contact with the latter surfaces in a special area called the **landing zone**. The landing zone is either the innermost or outermost cylinder on the drive.

When the drive is running and the patters are spinning, the heads float slightly above the surface of the platters on a cushion of air. The distance between the heads and the platters is called the head gap or the floating height.

**Heads**

- Ferrite Heads
- Metal-in-Gap Heads
- Thin Film Heads
- Magneto-Resistive Heads

**Ferrite Heads**

The oldest and the simplest head.  A U-shaped iron core wrapped with electrical windings. When writing, the current in the coil creates a polarized magnetic field in the gap between the poles of the core. When reading, the head is passed over the magnetic fields and a current is induced in the windings. the drive produces a magnetic field by energizing the coils or passing a magnetic field near them. This gives the heads full read/write capability. Ferrite heads are larger and heavier than thin-film heads and therefore require a larger floating height to prevent contact with the disk while it is spinning.

**Metal-in-Gap Heads**

Same design as ferrite heads, but add a metallic alloy in the head. Greatly increase its magnetization capabilities, allowing MIG heads to be used with higher density media.MIG heads are a specially enhanced version of the composite ferrite design. In MIG heads, a metal substance is applied to the head's recording gap.

### Thin Film Heads

Thin-film heads are manufactured much the same way as a semiconductor chip—through a photolithographic process. This process creates many thousands of heads on a single circular wafer and produces a small, high-quality product. TF heads have an extremely narrow and controlled head gap that is created by sputtering a hard aluminum material. Because this material completely encloses the gap, the area is well protected, minimizing the chance of damage from contact with the spinning disk. The core is a combination of iron and nickel alloy that has two to four times more magnetic power than a ferrite head core.

### Magneto-Resistive Heads

An MR head employs a special conductive material that changes its resistance in the presence of a magnetic field. As the head passes over the surface of the disk, this material changes resistance as the magnetic fields change corresponding to the stored patterns on the disk. A sensor is used to detect these changes in resistance, which allows the bits on the platter to be read. The use of MR heads allows much higher areal densities to be used on the platters than is possible with older designs, greatly increasing the storage capacity.

## Head Actuator

An actuator is the component that moves a hard drive's heads over the media surface, to read and write data. Each recording head sits at the end of a moving actuator arm. Hard drives today are equipped with a single actuator, which moves all the read-write heads together in synchronous motion.

### Hard disk Head Actuator Mechanisms

This mechanism moves the heads across the disk and positions them accurately above the desired cylinder. Many variations on head actuator mechanisms are in use, but all fall into one of two basic categories:

- Stepper motor actuators
- Voice coil actuators

### Stepper Motor Actuators

A stepper motor is an electrical motor that can "step," or move from position to position, with mechanical detents or click-stop positions. Stepper motors can't position themselves between step positions; they can stop only at the predetermined detent positions.

### Voice Coil Actuators

A voice coil actuator works by pure electromagnetic force. The construction of the mechanism is similar to that of a typical audio speaker, from which the term voice coil is derived. In a typical hard disk drive's voice coil system, the electromagnetic coil is attached to the end of the head rack and placed near a stationary magnet.

No physical contact occurs between the coil and the magnet; instead, the coil moves by pure magnetic force. As the electromagnetic coils are energized, they attract or repulse the stationary magnet and move the head rack. Systems like these are extremely quick, efficient, and usually much quieter than systems driven by stepper motors. Unlike a stepper motor, a voice coil actuator has no click-stops or detent positions; rather, a special guidance system stops the head rack above a particular cylinder. Because it has no detents, the voice coil actuator can slide the heads in and out smoothly to any position desired. Voice coil actuators use a guidance mechanism called a *servo* to tell the actuator where the heads are in relation to the cylinders and to place the heads accurately at the desired positions. This positioning system often is called a closed loop feedback mechanism.

### Spindle Motor

A *spindle motor* is a small, high-precision, high-reliability electric motor that is used to rotate the shaft, or *spindle*, on which the *platters* are mounted in a hard disk drive (HDD). Modern HDDs typically contain multiple platters, all of which are mounted on a single shaft, in order to maximize the data storage surface in a given volume of space. Most HDD spindle motors spin the platters at a constant rate ranging from 3,600 to 7,200 RPM.

### Logic Board

Hard disk is made with an intelligent circuit board integrated into the hard disk unit. It is mounted on the bottom of the base casting exposed to the outer side. The read/write heads are linked to the logic board through a flexible ribbon cable.

## Hard Drive Specification

- **Seek Time** is measured defines the amount of time it takes a hard drive's read/write head to find the physical location of a piece of data on the disk.
- **Latency** is the average time for the sector being accessed to rotate into position under a head, after a completed seek.
- **Access time** is the time delay or latency between the beginning of a read or write operation and the time when the drive actually begins reading or writing data.

# Optical storage Devices

In the **optical storage devices**, all read and write activities are performed by light. All recording information stores at an optical disk. As per the opinions of data scientist that compact space is most useful for huge data storage. Their big advantages are not more costly, light weight, and easy to transport because it is removable device unlike hard drive.

- Eg: **CD**-ROM,**DVD-ROM**, **Blue Ray**,

### Optical Storage Media

Like magnetic devices, optical storage is digital; the drive reads a sequence of ones and zeroes from the disk and interprets it into the data used by the PC. The head on a CD-ROM drive consists of a light source and a light sensor. The drive bounces a beam of light off of the surface of the disk and the sensor reads the reflect light. The surface of the disc consists of smooth areas called lands and rough areas called pits. When the beam of light bounces off of a land, it's reflected back to the sensor at nearly its full strength. When the beam bounces off of a pit, the light is diffused, and arrives at the sensor with much less strength. The difference in the strength of the light beams reaching the sensor forms a pattern used to encode the binary data, just as magnetic media use a pattern of magnetic flux reversals.

## Compact Discs

Abbreviated as CD, a compact disc is a flat, round, optical storage medium. compact discs store data to be retrieved or executed at a later date. CDs can store software programs to install onto your computer. They save files for backup or transfer to another computer, and hold music to play in a CD player.The standard CD is capable of holding 72 minutes of music or 650 MB of data. An 80 minute CD is capable of holding 700 MB of data.

## How do CD-R discs work

The surface of the CD contains one long spiral track of data. Along the track, there are flat reflective areas and non-reflective bumps. A flat reflective area represents a binary 1, while a non-reflective bump represents a binary 0. The CD drive shines a laser at the surface of the CD and can detect the reflective areas and the bumps by the amount of laser light they reflect. The drive converts the reflections into 1s and 0s to read digital data from the disc.Normal CDs can not be modified -- they are read-only devices.

## CD-R disc

A CD-R disc needs to allow the drive to write data onto the disc. For a CD-R disk to work, there must be a way for a laser to create a non-reflective area on the disc. A CD-R disc therefore has an extra layer that the laser can modify. This extra layer is a greenish dye. In a normal CD, you have a plastic substrate covered with a reflective aluminum or gold layer. In a CD-R, you have a plastic substrate, a dye layer and a reflective gold layer. On a new CD-R disc, the entire surface of the disc is reflective -- the laser can shine through the dye and reflect off the gold layer. When you write data to a CD-R, the writing laser (which is much more powerful than the reading laser) **heats up** the dye layer and changes its transparency. The change in the dye creates the equivalent of a non-reflective bump.

## Compact Disc Formats

- **Red Book:** known as the Compact Disc Digital Audio or Audio CD format. It is the original and oldest Compact Disc standard and the foundation for all other standards. It is an audio-only format used on every Audio CD. The audio on these discs is usually referred to as Red Book audio or CD-quality audio. Up to 74 minutes of sound on a disc.

- **Yellow Book:** known as Compact Disc-Read-Only Memory. The introduction of this architecture allowed Compact Discs to be used as an archival medium for computer data. Used to store up to 650 MB of data.
- **Green Book:** The architecture defined in the Green Book helped to improve the synchronization of data retrieval and audio information and established the Compact Disc Interactive format.
- **Orange Book:** the major contribution of the Orange book to CD-ROM is its foundation for CD-R technology. In addition, this architecture allows multiple sessions to be recorded on a single disc.
- **White Book:** to store movies and high-quality video presentations. This standard is based on CD-ROM XA. It uses MPEG-1 to compress audio and video.
- **Blue Book:** A multi-session format, The first session contains a regular selection of audio tracks, while the second session contains computer data and/or video clips. Discs using this standard can be played in a normal CD player as standard audio discs and in a computer CD-ROM player as multimedia discs.

## DVDs

Stands for "Digital Versatile Disc." A DVD is a type of optical media used for storing digital data. It is the same size as a CD, but has a larger storage capacity. Some DVDs are formatted specifically for video playback, while others may contain different types of data, such as software programs and computer files. A standard video DVD can store 4.7 GB of data, which is enough to hold over 2 hours of video in 720p resolution, using MPEG-2 compression. DVDs are also used to distribute software programs. Since some applications and other software (such as clip art collections) are too large to fit on a single 700 MB CD, DVDs provide a way to distribute large programs on a single disc. Writable DVDs also provide a way to store a large number of files and back up data. The writable DVD formats include DVD-R, DVD+R, DVD-RW, DVD+RW, and DVD-RAM. a dual-layer DVD (which has two layers of data on a single side of the disc) can store 8.5 GB of data. A dual-sided DVD can store 9.4 GB of data (4.7 x 2). A dual-layer, dual-sided DVD can store 17.1 GB of data.

**DVD formats**

- DVD-ROM – Read Only digital data storage
- DVD Video – Digital video storage for movies and other long format media.
- DVD Audio – Digital audio storage
- DVD-R – Write once, read many
- DVD-RAM – Rewritable digital data storage

**Advantages of DVD**

- The picture quality is far better than a CD.

- Most DVD's have Dolby Digital or DTS. Thus the clarity of the sound will be nearly equal to that in a theatre.
- The DVD has a special on-screen index quality which helps you to take a closer look at important scenes of a movie which was earlier indexed by the publisher of the movie.
- Unlike a CD player, the DVD player is specialized in taking you to the correct part which you prefer to see. Thus, there will be no need of fast-forwarding.
- The format of a DVD can be changed according to the type of view you need. It can be a standard TV size format and also a wide-screen TV format.

**How DVD work**

DVDs are of the same diameter and thickness as CDs, and they are made using some of the same materials and manufacturing methods. Like a CD, the data on a DVD is encoded in the form of small pits and bumps in the track of the disc. The manufacturing methods and materials used for manufacture as the same as that of a CD. The DVD has a number of layers of plastic pieces with a total thickness of 1.2 millimetres. Clear poly-carbonate plastic is basically used for making these layers.

**DVD-ROM Drives**

DVD ROM drives are similar in construction and operation to CD ROM drives.CD/DVD drives are used to read data and applications from CDs (compact disk) and DVDs (digital versatile disk). It is a device which uses laser technology to read and write data on/from optical disks. DVD drives also spin the disc at far slower rate than do CD-ROM drives, but because the data is more densely packed onto the disc, the transfer rate of a DVD is much higher. While a single speed CD-ROM can read data at 150 KB/sec, a single speed DVD-ROM drive can transfer data at 1,250 KB/sec. Data is written on optical disk by burning pits on the disk with laser. Later on, data can be read by reflecting low-power laser onto the surface of the disk. The photo detector then translates the bouncing lights back into sound or data.

# Recordable drives

- Some technologies can write to the same disk many times.
- CD-R
- The CD-R Disk
- The CD-R Drive
- CD-R Software
- CD-RW

## CD-R

Stands for "Compact Disc Recordable." CD-R discs are blank CDs that can record data written by a CD burner. The word "recordable" is used because CD-Rs are often used to record audio, which can be played back by most CD players. However, many other kinds of data can also be written to a CD-R, so the discs are also referred to as "writable CDs.“ The data burned onto a CD-R disc is permanent, meaning it can not be altered or erased like the data on a hard drive. Typically, once a CD has been burned, it will not be able to record any more data. Some CD burning programs can record data as "sessions," allowing a disc to be written to mulitple times until it is full. Each session creates a new partition on the disc, meaning a computer will read a disc with multiple sessions as multiple discs.

## The CD-R Disc

- A CD-R (Compact Disc Recordable) is a type of optical disc that can be written to, or recorded, only once. CD-Rs are part of the CD family of discs, which also includes CD-ROMs (Read-Only Memory) and CD-RWs (Rewritable).
- A CD-R disc has a dye layer on its top side, which is sandwiched between the polycarbonate substrate (base) and the reflective layer.
- The reflective layer typically consists of aluminum. The dye layer is where the data is physically recorded during the writing process.
- CD-Rs are recorded using a laser to alter the physical characteristics of the dye layer.
- During the writing process, the laser heats the dye layer, causing a chemical change. This change makes certain areas of the dye layer more or less reflective.

## The CD-R Drive

The CD-R drive itself contains most of the same components as a regular drive, except for the special laser. CD-R drives generally have two speed ratings, one for reading discs and one for writing them. The writing speed inevitably is lower than the reading speed. CD-R drives are not as fast as standard CD-ROM drives. CD-R drives are available in both IDE and SCSI versions.

## CD-R Software

In addition to the drive and the blank discs, to burn CDs on your computer, it need special software, which is nearly always included with the drive. This software provides an interface that you can use to select the files that you want to write to the disc. User can usually select files and directories located anywhere on your computer and organize the directory structure of the disc however you wish. Sometimes, the software even enables you to select files on other computer connected by a network.

**Single-Session and Multisession Discs**

A session is a continuously written collection of data written to a CD-ROM. Commercially produced CDs are all single-session discs, because all of their data is written at one time. It is possible to write multiple session on a disc. It need a multisession CD-ROM drive to read them.

## CD-RW

Stands for "Compact Disc Re-Writable." A CD-RW is a blank CD that can be written to by a CD burner. Unlike a CD-R (CD-Recordable), a CD-RW can be written to multiple times. The data burned on a CD-RW cannot be changed, but it can be erased. Therefore, you have to completely erase a CD-RW every time you want to change the files or add new data. While it may be somewhat inconvenient, this capability makes CD-RWs a good choice for making frequent backups.

# Solid-state drive (SSD)

A solid-state drive (SSD) is a nonvolatile storage device that stores persistent data on solid-state flash memory. A traditional hard disk drive (HDD) consists of a spinning disk with a read/write head on a mechanical arm called an actuator. An SSD, on the other hand, has an array of semiconductor memory organized as a disk drive, using integrated circuits (ICs) rather than magnetic or optical storage media.

### *SSD provides three distinct advantages:*

- Less Power Usage
- Faster Data Access
- Higher Reliability

## SSD vs. HDD pros and cons

SSD performance is considered to be much faster than the highest performance electromechanical disk drives. Seek time and latency are also substantially reduced, and end users typically enjoy much faster boot times.

In general, SSDs are more durable and much quieter than HDDs, with no moving parts to break or spin up or down. SSDs employ wear leveling to increase drive lifespan.

Wear leveling is typically managed by the flash controller, which uses an algorithm to arrange data so write/erase cycles are distributed evenly among all the blocks in the device.

| | SSD | HDD |
|---|---|---|
| **Access time** | An SSD has access speeds of 35 to 100 micro-seconds, which is nearly 100 times faster. | A typical HDD takes about 5,000 to 10,000 micro-seconds to access data. |
| **Reliability** | The SSD drive has no moving parts. It uses flash memory to store data, which provides better performance and reliability over an HDD | The HDD has moving parts and magnetic platters, meaning the more use they get, the faster they wear down and fail. |
| **Power** | The SSD uses less power than a standard HDD, which means a lower energy bill over time and for laptops an increase of battery life | With all the parts and requirements to spin the platters, the HDD uses more power than an SSD |